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HAZARDOUS WASTE & RCRA

Understanding Hazardous Waste in Copper Refining



What is Hazardous Waste?

Hazardous waste is any discarded material that can harm human health or the environment. Copper refining may generate hazardous wastes such as acidic sludges, metal-containing residues, and spent electrolyte solutions, which must be managed in accordance with federal regulations.



RCRA Hazardous Waste

Characteristics



Ignitability

Flash point $< 140^{\circ}\text{F}$ (60°C)



Corrosivity



Reactivity

May react violently or release toxic gases



Toxicity

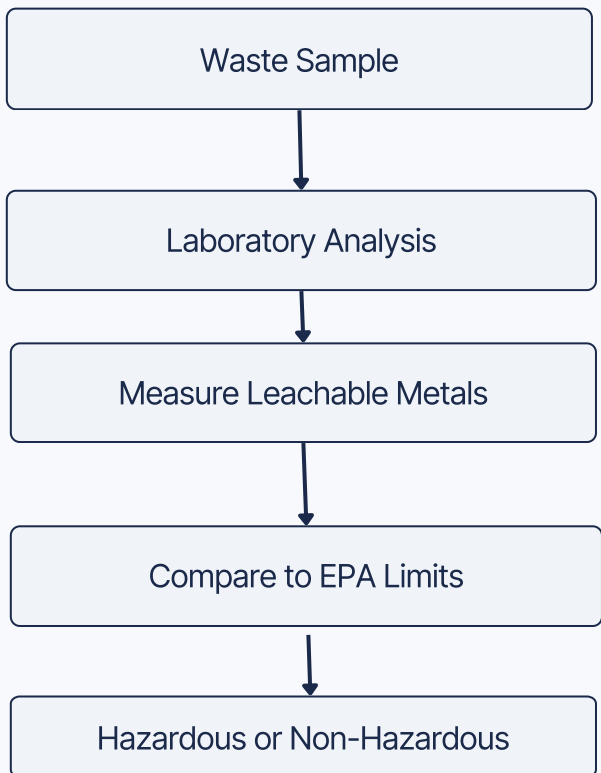
Exceeds EPA TCLP regulatory limits

HAZARDOUS WASTE & RCRA



What is TCLP?

The Toxicity Characteristic Leaching Procedure (TCLP, EPA Method 1311) determines whether hazardous constituents can leach from a waste under landfill conditions. Results are compared with EPA regulatory limits to determine if the waste is hazardous.



HAZARDOUS WASTE & RCRA



TCLP Regulatory Limits

As

Arsenic

5.0 mg/L

Ba

Barium

100 mg/L

Cd

Cadmium

1.0 mg/L

Cr

Chromium

5.0 mg/L

Pb

Lead

5.0 mg/L

Hg

Mercury

0.2 mg/L

Se

Selenium

1.0 mg/L

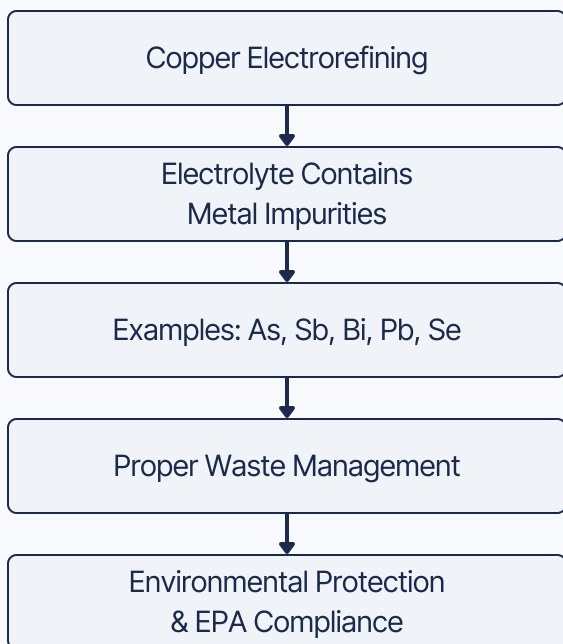
Ag

Silver

5.0 mg/L

COPPER ELECTROREFINING

Why is this Important in Copper Electrowinning?



Important Note

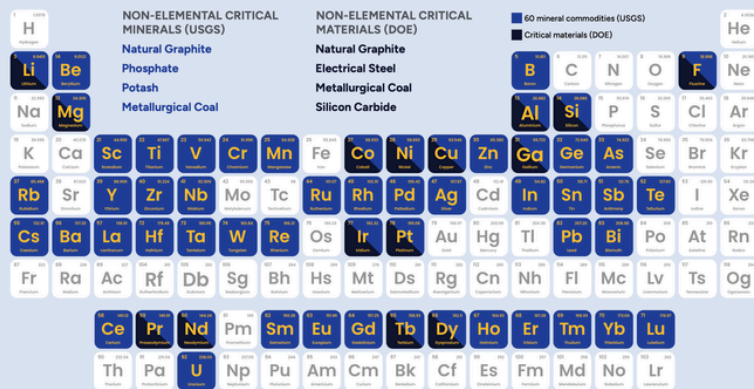
Note: Antimony (Sb) and bismuth (Bi) are not RCRA 8 metals; however, they are important impurities monitored in copper electrorefining and were the focus of this research.

CRITICAL MINERALS & MATERIALS

Why Are Sb and Bi Important?

Antimony (Sb) and bismuth (Bi) are recognized as critical minerals because they are important for advanced manufacturing, clean energy technologies, and national security. Recovering these elements from copper electrorefining electrolyte can improve resource efficiency while supporting a more secure domestic supply of these valuable materials.

THE ELEMENTS OF CRITICAL MINERALS AND MATERIALS



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